Summer Internship Report

AWS Data Engineering

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Internship Certificate

This is to certify that **BHANU UPENDRA HARSHA KUMAR PULIVARTHI** from Prasad V. Potluri Siddhartha Institute of Technology has successfully completed 10 weeks **Data Engineering Virtual Internship** during January - March 2025.

The internship was conducted by AICTE-EduSkills Foundation, with curriculum provided by AWS Academy.

Learning Objectives

- Understand fundamental concepts of data engineering and its role in analytics and ML.
- Gain hands-on experience with AWS tools like S3, Glue, Athena, and Redshift.
- Develop skills to build efficient and scalable data pipelines tailored to specific applications.
- Explore real-world use cases, data collection strategies, and security best practices.
- Familiarize with Amazon SageMaker for machine learning model integration.
- Enhance decision-making capabilities based on data insights and analysis.

Abstract

"This internship on AWS Data Engineering provides comprehensive knowledge on how data engineering plays a crucial role in the modern data-driven landscape. It introduces key concepts such as collecting, storing, and preparing data for analytics and machine learning (ML) applications... The report outlines the practical modules covered, including the development of secure, scalable, and efficient data pipelines, as well as hands-on experience with AWS tools."

Introduction to Topic & Domain

Domain: Cloud & Data Engineering

The domain focuses on using cloud platforms to build robust data infrastructure. This is essential for modern, data-driven organizations.

Topic: AWS Data Engineering

This involves using the suite of Amazon Web Services to design, build, and maintain systems that collect, store, process, and analyze data at scale. It forms the backbone for analytics and machine learning.

What is a Data Pipeline?

A series of automated steps that move and transform data from a source (like an app or database) to a destination (like a data warehouse) for analysis.

Core Modules Overview: Weeks 1-5 (Fundamentals)

1. Intro & Role

Defined the DE role, org needs, and data pipelines.

2. 5 V's & S3

Explored the 5 V's of data and Amazon S3 storage.

3. Architectures

Evolution of data systems to modern data lakes on AWS.

4. Security & Scale

Cloud security fundamentals and scalable infrastructure.

5. ETL vs. ELT

Data ingestion, wrangling, cleaning, and validation.

Core Modules Overview: Weeks 6-10 (Advanced Processing)

6. Redshift DW

Data lakes vs. warehouses, deep dive into Amazon Redshift.

7. Big Data & EMR

Processing Big Data using Hadoop, Spark, and Amazon EMR.

8. ML & SageMaker

ML lifecycle and processing data for Amazon SageMaker.

9. Analytics & Kinesis

Data analysis with Athena, QuickSight, and Kinesis streaming.

10. Automation

Automating infrastructure and workflows using Step Functions.

Conclusion & Key Takeaways

The 10-week internship covered the complete **Data Engineering Lifecycle**, from mastering scalable storage (S3/Redshift) to advanced processing (EMR/SageMaker) and pipeline automation (Step Functions).

This experience culminated in the ability to design secure, efficient, and fully automated ETL/ELT solutions on AWS, providing essential skills for real-world data infrastructure challenges.

Thank You

Image Sources

https://i.ytimg.com/vi/-kPZS_AdzA0/hq720.jpg?sqp=-oaymwEhCK4FEIIDSFryq4qpAxMIARUAAAAAGAEIAADIQj0AgKJD&rs=AOn4CLC9LYnu3UH86e_n3xsmIfgLhppfrw

Source: www.youtube.com

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